Logistic_Reg

October 23, 2026

[8]:
```
import pandas as pd
import numpy as np
import pickle
```

[9]:
```
dataset = pd.read_csv("Social_Network_Ads.csv")
dataset
```

[9]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	…	…	…	…	…
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

[10]:
```
dataset = pd.get_dummies(dataset,drop_first=True)
dataset
```

[10]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
..	…	…	…	…	…
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

```
[400 rows x 5 columns]
```

[11]:
```
dataset = dataset.drop("User ID",axis=1)
dataset
```

[11]:
```
      Age  EstimatedSalary  Purchased  Gender_Male
0      19            19000          0         True
1      35            20000          0         True
2      26            43000          0        False
3      27            57000          0        False
4      19            76000          0         True
..    …               …        …           …
395    46            41000          1        False
396    51            23000          1         True
397    50            20000          1        False
398    36            33000          0         True
399    49            36000          1        False

[400 rows x 4 columns]
```

[12]:
```
dataset["Purchased"].value_counts()
```

[12]:
```
Purchased
0    257
1    143
Name: count, dtype: int64
```

[14]:
```
dataset.shape
```

[14]:
```
(400, 4)
```

[15]:
```
x=dataset[["Gender_Male","Age","EstimatedSalary"]]
y=dataset[["Purchased"]]
```

[16]:
```
y
```

[16]:
```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         …
395          1
396          1
397          1
398          0
399          1
```

```
[400 rows x 1 columns]
```

[17]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 42 )
```

[19]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

[41]:
```
clf = LogisticRegression(random_state=None,verbose=0,n_jobs=None,C=1.0)
clf.fit(x_train,y_train)
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:1368: DataConversionWarning: A column-
vector y was passed when a 1d array was expected. Please change the shape of y
to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[41]:
```
LogisticRegression()
```

[42]:
```
y_pred = clf.predict(x_test)
y_pred
```

[42]:
```
array([0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0,
       0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0,
       1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0])
```

[43]:
```
cm = confusion_matrix(y_test,y_pred)
cm
```

[43]:
```
array([[50,  2],
       [ 7, 21]])
```

[39]:
```
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.85      0.96      0.90        52
           1       0.90      0.68      0.78        28

    accuracy                           0.86        80
   macro avg       0.88      0.82      0.84        80
weighted avg       0.87      0.86      0.86        80
```

[ ]:

KN_Neighbors

October 23, 2026

[44]:
```
import pandas as pd
import numpy as np
import pickle
```

[45]:
```
dataset = pd.read_csv("Social_Network_Ads.csv")
dataset
```

[45]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	…	…	…	…	…
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

[46]:
```
dataset = pd.get_dummies(dataset,drop_first=True)
dataset
```

[46]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
..	…	…	…	…	…
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

```
[400 rows x 5 columns]
```

[47]:
```
dataset = dataset.drop("User ID",axis=1)
dataset
```

[47]:
```
      Age  EstimatedSalary  Purchased  Gender_Male
0      19            19000          0         True
1      35            20000          0         True
2      26            43000          0        False
3      27            57000          0        False
4      19            76000          0         True
..    …               …        …           …
395    46            41000          1        False
396    51            23000          1         True
397    50            20000          1        False
398    36            33000          0         True
399    49            36000          1        False

[400 rows x 4 columns]
```

[48]:
```
dataset["Purchased"].value_counts()
```

[48]:
```
Purchased
0    257
1    143
Name: count, dtype: int64
```

[49]:
```
dataset.shape
```

[49]:
```
(400, 4)
```

[50]:
```
x=dataset[["Gender_Male","Age","EstimatedSalary"]]
y=dataset[["Purchased"]]
```

[51]:
```
y
```

[51]:
```
      Purchased
0             0
1             0
2             0
3             0
4             0
..          …
395           1
396           1
397           1
398           0
399           1
```

```
[400 rows x 1 columns]
```

[52]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 42 )
```

[53]:
```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

[54]:
```
clf =␣
 ↪KNeighborsClassifier(n_neighbors=5,weights='uniform',algorithm='auto',n_jobs=None)
clf.fit(x_train,y_train)
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\neighbors\_classification.py:243: DataConversionWarning: A
column-vector y was passed when a 1d array was expected. Please change the shape
of y to (n_samples,), for example using ravel().
  return self._fit(X, y)
```

[54]:
```
KNeighborsClassifier()
```

[55]:
```
y_pred = clf.predict(x_test)
y_pred
```

[55]:
```
array([1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0,
       1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1,
       1, 1, 0, 1, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0])
```

[56]:
```
cm = confusion_matrix(y_test,y_pred)
cm
```

[56]:
```
array([[46,  6],
       [ 8, 20]])
```

[57]:
```
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.85      0.88      0.87        52
           1       0.77      0.71      0.74        28

    accuracy                           0.82        80
   macro avg       0.81      0.80      0.80        80
weighted avg       0.82      0.82      0.82        80
```

[ ]:

Gausian_Naives

October 23, 2026

[58]:
```
import pandas as pd
import numpy as np
import pickle
```

[59]:
```
dataset = pd.read_csv("Social_Network_Ads.csv")
dataset
```

[59]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	…	…	…	…	…
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

[60]:
```
dataset = pd.get_dummies(dataset,drop_first=True)
dataset
```

[60]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
..	…	…	…	…	…
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

```
[400 rows x 5 columns]
```

[61]:
```
dataset = dataset.drop("User ID",axis=1)
dataset
```

[61]:
```
      Age  EstimatedSalary  Purchased  Gender_Male
0      19            19000          0         True
1      35            20000          0         True
2      26            43000          0        False
3      27            57000          0        False
4      19            76000          0         True
..    …               …        …           …
395    46            41000          1        False
396    51            23000          1         True
397    50            20000          1        False
398    36            33000          0         True
399    49            36000          1        False

[400 rows x 4 columns]
```

[62]:
```
dataset["Purchased"].value_counts()
```

[62]:
```
Purchased
0    257
1    143
Name: count, dtype: int64
```

[63]:
```
dataset.shape
```

[63]:
```
(400, 4)
```

[64]:
```
x=dataset[["Gender_Male","Age","EstimatedSalary"]]
y=dataset[["Purchased"]]
```

[65]:
```
y
```

[65]:
```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         …
395          1
396          1
397          1
398          0
399          1
```

```
[400 rows x 1 columns]
```

[66]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 42 )
```

[67]:
```
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

[68]:
```
clf = GaussianNB(priors=None,var_smoothing=1e-09)
clf.fit(x_train,y_train)
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:1368: DataConversionWarning: A column-
vector y was passed when a 1d array was expected. Please change the shape of y
to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[68]:
```
GaussianNB()
```

[69]:
```
y_pred = clf.predict(x_test)
y_pred
```

[69]:
```
array([1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0,
       1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1,
       0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1,
       1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0])
```

[70]:
```
cm = confusion_matrix(y_test,y_pred)
cm
```

[70]:
```
array([[50,  2],
       [ 3, 25]])
```

[71]:
```
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.94      0.96      0.95        52
           1       0.93      0.89      0.91        28

    accuracy                           0.94        80
   macro avg       0.93      0.93      0.93        80
weighted avg       0.94      0.94      0.94        80
```

[ ]:

Multinomial

October 23, 2026

[72]:
```
import pandas as pd
import numpy as np
import pickle
```

[73]:
```
dataset = pd.read_csv("Social_Network_Ads.csv")
dataset
```

[73]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	…	…	…	…	…
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

[74]:
```
dataset = pd.get_dummies(dataset,drop_first=True)
dataset
```

[74]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
..	…	…	…	…	…
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

```
[400 rows x 5 columns]
```

[75]:
```
dataset = dataset.drop("User ID",axis=1)
dataset
```

[75]:
```
      Age  EstimatedSalary  Purchased  Gender_Male
0      19            19000          0         True
1      35            20000          0         True
2      26            43000          0        False
3      27            57000          0        False
4      19            76000          0         True
..    …               …        …           …
395    46            41000          1        False
396    51            23000          1         True
397    50            20000          1        False
398    36            33000          0         True
399    49            36000          1        False

[400 rows x 4 columns]
```

[76]:
```
dataset["Purchased"].value_counts()
```

[76]:
```
Purchased
0    257
1    143
Name: count, dtype: int64
```

[77]:
```
dataset.shape
```

[77]:
```
(400, 4)
```

[78]:
```
x=dataset[["Gender_Male","Age","EstimatedSalary"]]
y=dataset[["Purchased"]]
```

[79]:
```
y
```

[79]:
```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         …
395          1
396          1
397          1
398          0
399          1
```

```
[400 rows x 1 columns]
```

[80]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 42 )
```

[81]:
```
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

[82]:
```
clf = MultinomialNB(alpha=1.0,force_alpha=True,fit_prior=True,class_prior=None)
clf.fit(x_train,y_train)
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:1368: DataConversionWarning: A column-
vector y was passed when a 1d array was expected. Please change the shape of y
to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[82]:
```
MultinomialNB()
```

[83]:
```
y_pred = clf.predict(x_test)
y_pred
```

[83]:
```
array([0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1,
       0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0,
       0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1,
       1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0])
```

[84]:
```
cm = confusion_matrix(y_test,y_pred)
cm
```

[84]:
```
array([[38, 14],
       [18, 10]])
```

[85]:
```
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.68      0.73      0.70        52
           1       0.42      0.36      0.38        28

    accuracy                           0.60        80
   macro avg       0.55      0.54      0.54        80
weighted avg       0.59      0.60      0.59        80
```

[ ]:

Bernauli

October 23, 2026

[86]:
```
import pandas as pd
import numpy as np
import pickle
```

[87]:
```
dataset = pd.read_csv("Social_Network_Ads.csv")
dataset
```

[87]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	…	…	…	…	…
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

[88]:
```
dataset = pd.get_dummies(dataset,drop_first=True)
dataset
```

[88]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
..	…	…	…	…	…
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

```
[400 rows x 5 columns]
```

[89]:
```
dataset = dataset.drop("User ID",axis=1)
dataset
```

[89]:
```
      Age  EstimatedSalary  Purchased  Gender_Male
0      19            19000          0         True
1      35            20000          0         True
2      26            43000          0        False
3      27            57000          0        False
4      19            76000          0         True
..    …               …        …           …
395    46            41000          1        False
396    51            23000          1         True
397    50            20000          1        False
398    36            33000          0         True
399    49            36000          1        False

[400 rows x 4 columns]
```

[90]:
```
dataset["Purchased"].value_counts()
```

[90]:
```
Purchased
0    257
1    143
Name: count, dtype: int64
```

[91]:
```
dataset.shape
```

[91]:
```
(400, 4)
```

[92]:
```
x=dataset[["Gender_Male","Age","EstimatedSalary"]]
y=dataset[["Purchased"]]
```

[93]:
```
y
```

[93]:
```
      Purchased
0             0
1             0
2             0
3             0
4             0
..          …
395           1
396           1
397           1
398           0
399           1
```

```
[400 rows x 1 columns]
```

[94]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 42 )
```

[95]:
```
from sklearn.naive_bayes import BernoulliNB
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

[96]:
```
clf =BernoulliNB(alpha=1.0,force_alpha=True,fit_prior=True,binarize=0.
 ↪0,class_prior=None)
clf.fit(x_train,y_train)
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:1368: DataConversionWarning: A column-
vector y was passed when a 1d array was expected. Please change the shape of y
to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[96]:
```
BernoulliNB()
```

[97]:
```
y_pred = clf.predict(x_test)
y_pred
```

[97]:
```
array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0])
```

[98]:
```
cm = confusion_matrix(y_test,y_pred)
cm
```

[98]:
```
array([[52,  0],
       [28,  0]])
```

[99]:
```
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.65      1.00      0.79        52
           1       0.00      0.00      0.00        28

    accuracy                           0.65        80
   macro avg       0.33      0.50      0.39        80
weighted avg       0.42      0.65      0.51        80
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\metrics\_classification.py:1879: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\metrics\_classification.py:1879: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\metrics\_classification.py:1879: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```

Categorical

October 23, 2026

[86]:
```
import pandas as pd
import numpy as np
import pickle
```

[87]:
```
dataset = pd.read_csv("Social_Network_Ads.csv")
dataset
```

[87]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	…	…	…	…	…
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

[88]:
```
dataset = pd.get_dummies(dataset,drop_first=True)
dataset
```

[88]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
..	…	…	…	…	…
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

```
[400 rows x 5 columns]
```

[89]:
```
dataset = dataset.drop("User ID",axis=1)
dataset
```

[89]:
```
      Age  EstimatedSalary  Purchased  Gender_Male
0      19            19000          0         True
1      35            20000          0         True
2      26            43000          0        False
3      27            57000          0        False
4      19            76000          0         True
..    …               …        …           …
395    46            41000          1        False
396    51            23000          1         True
397    50            20000          1        False
398    36            33000          0         True
399    49            36000          1        False

[400 rows x 4 columns]
```

[90]:
```
dataset["Purchased"].value_counts()
```

[90]:
```
Purchased
0    257
1    143
Name: count, dtype: int64
```

[91]:
```
dataset.shape
```

[91]:
```
(400, 4)
```

[92]:
```
x=dataset[["Gender_Male","Age","EstimatedSalary"]]
y=dataset[["Purchased"]]
```

[93]:
```
y
```

[93]:
```
      Purchased
0             0
1             0
2             0
3             0
4             0
..          …
395           1
396           1
397           1
398           0
399           1
```

```
[400 rows x 1 columns]
```

[100]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 42 )
```

[101]:
```
from sklearn.naive_bayes import CategoricalNB
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

[102]:
```
clf =CategoricalNB(alpha=1.
 ↪0,force_alpha=True,fit_prior=True,min_categories=None,class_prior=None)
clf.fit(x_train,y_train)
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:1368: DataConversionWarning: A column-
vector y was passed when a 1d array was expected. Please change the shape of y
to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[102]:
```
CategoricalNB()
```

[103]:
```
y_pred = clf.predict(x_test)
y_pred
```

[103]:
```
array([1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,
       1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1,
       0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0])
```

[104]:
```
cm = confusion_matrix(y_test,y_pred)
cm
```

[104]:
```
array([[50,  2],
       [12, 16]])
```

[105]:
```
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.81      0.96      0.88        52
           1       0.89      0.57      0.70        28

    accuracy                           0.82        80
   macro avg       0.85      0.77      0.79        80
weighted avg       0.84      0.82      0.81        80
```

[ ]:

Complement

October 23, 2026

[86]:
```
import pandas as pd
import numpy as np
import pickle
```

[87]:
```
dataset = pd.read_csv("Social_Network_Ads.csv")
dataset
```

[87]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	…	…	…	…	…
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

[88]:
```
dataset = pd.get_dummies(dataset,drop_first=True)
dataset
```

[88]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
..	…	…	…	…	…
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

```
[400 rows x 5 columns]
```

[89]:
```
dataset = dataset.drop("User ID",axis=1)
dataset
```

[89]:
```
      Age  EstimatedSalary  Purchased  Gender_Male
0      19            19000          0         True
1      35            20000          0         True
2      26            43000          0        False
3      27            57000          0        False
4      19            76000          0         True
..    …               …        …           …
395    46            41000          1        False
396    51            23000          1         True
397    50            20000          1        False
398    36            33000          0         True
399    49            36000          1        False

[400 rows x 4 columns]
```

[90]:
```
dataset["Purchased"].value_counts()
```

[90]:
```
Purchased
0    257
1    143
Name: count, dtype: int64
```

[91]:
```
dataset.shape
```

[91]:
```
(400, 4)
```

[92]:
```
x=dataset[["Gender_Male","Age","EstimatedSalary"]]
y=dataset[["Purchased"]]
```

[93]:
```
y
```

[93]:
```
      Purchased
0             0
1             0
2             0
3             0
4             0
..          …
395           1
396           1
397           1
398           0
399           1
```

```
[400 rows x 1 columns]
```

[132]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 24 )
```

[133]:
```
from sklearn.naive_bayes import ComplementNB
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

[127]:
```
clf =ComplementNB(alpha=1.
 ↪0,force_alpha=True,fit_prior=True,norm=False,class_prior=None)
clf.fit(x_train,y_train)
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:1368: DataConversionWarning: A column-
vector y was passed when a 1d array was expected. Please change the shape of y
to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[127]:
```
ComplementNB()
```

[128]:
```
y_pred = clf.predict(x_test)
y_pred
```

[128]:
```
array([1, 0, 1, 1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0,
       0, 0, 1, 0, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0,
       0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1,
       0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 0])
```

[129]:
```
cm = confusion_matrix(y_test,y_pred)
cm
```

[129]:
```
array([[22, 30],
       [18, 10]])
```

[130]:
```
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.55      0.42      0.48        52
           1       0.25      0.36      0.29        28

    accuracy                           0.40        80
   macro avg       0.40      0.39      0.39        80
weighted avg       0.45      0.40      0.41        80
```

[ ]: